ASE LAB 9

Perform Static Code Analysis of a Java Program Containing Duplicate Code Statements Using SonarQube to Identity Code of Duplication and Maintainability Issues.

Objective

To analyze a Java program using SonarQube and identify repeated or duplicate code using Copy-Paste Detection (CPD).

Step 1: Open Docker Desktop

Firstly, open Docker Desktop.

Run the SonarQube container using Command Prompt:

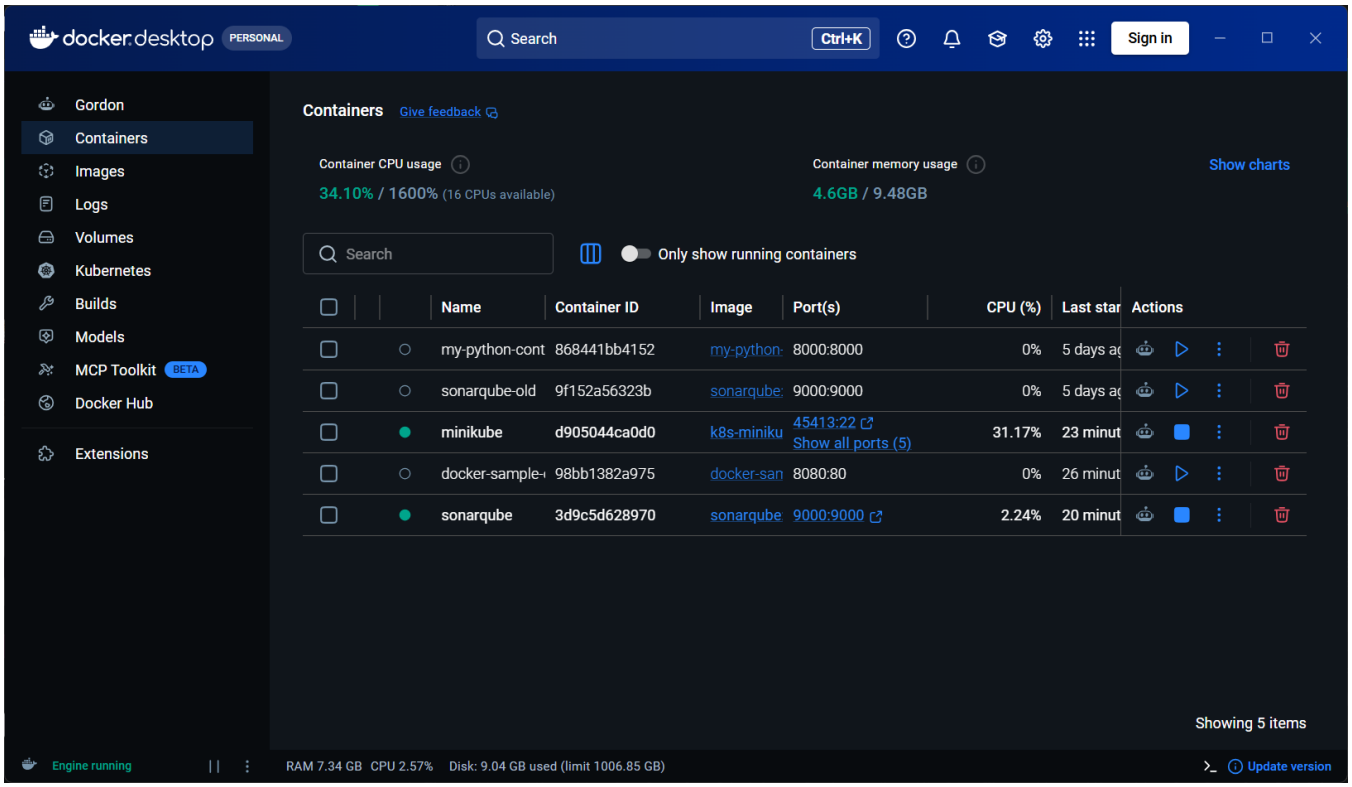
```
docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000
sonarqube:latest
```

Wait for the command to finish. Then check the running container:

```
docker ps
```

If the SonarQube container is shown in the output, the container is running successfully.

(On this system the SonarQube container was already running from Lab 8, so docker ps / Docker Desktop confirm it is up.)





Step 2: Open SonarQube using localhost

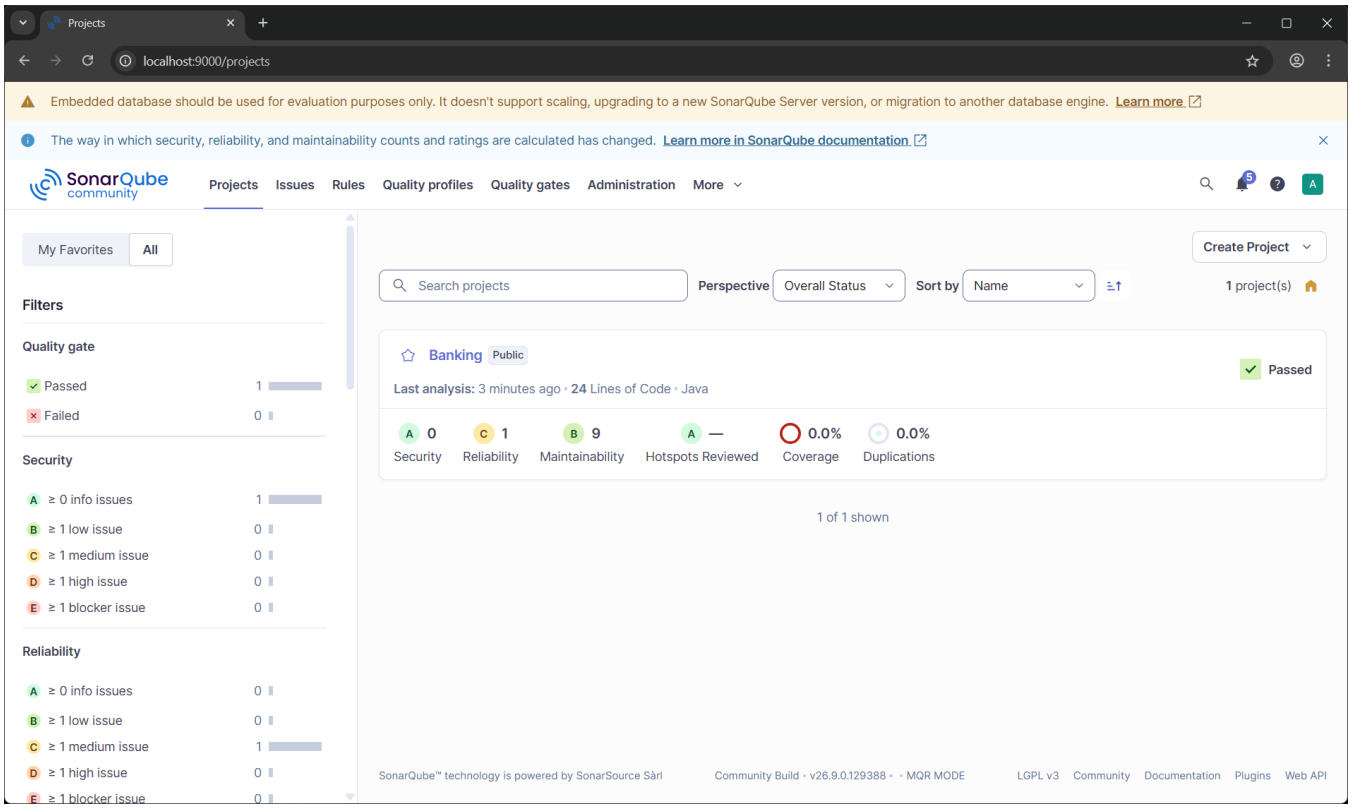
Open a browser and enter:

```
http://localhost:9000
```

Login using:

Username: admin

Password: admin



Step 3: Create the SonarQube Project

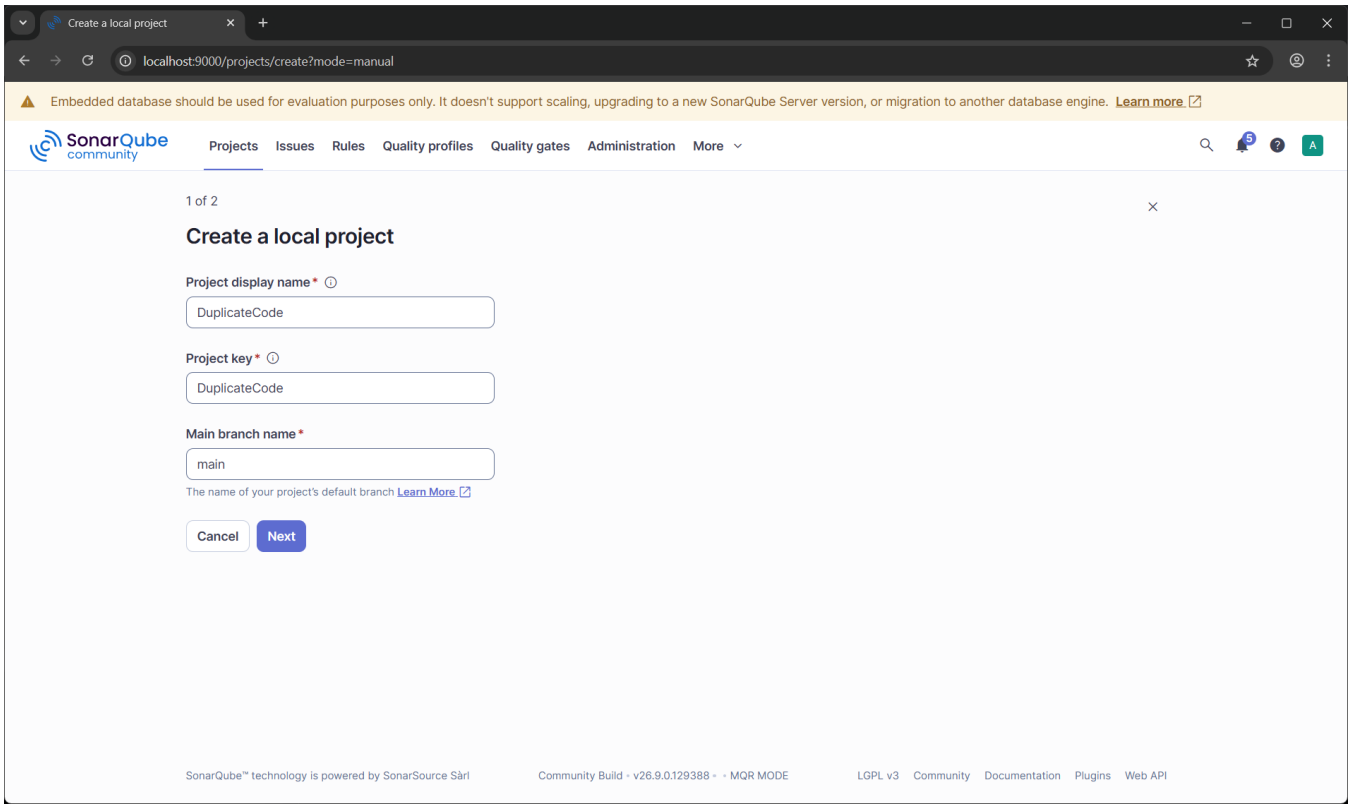
In SonarQube, create a new project.

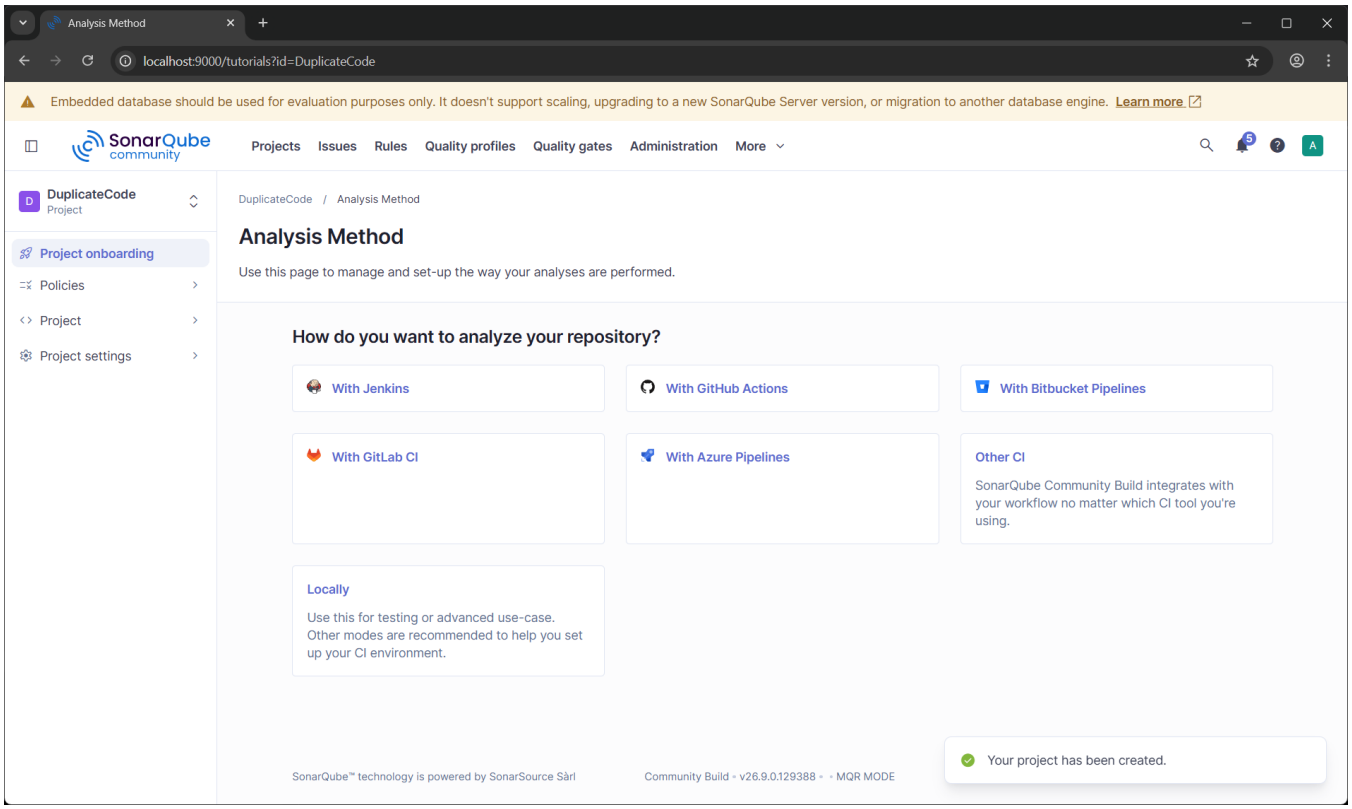
Enter:

Project Name: DuplicateCode

Project Key: DuplicateCode

Keep the project public if that is the setting used in your class setup.





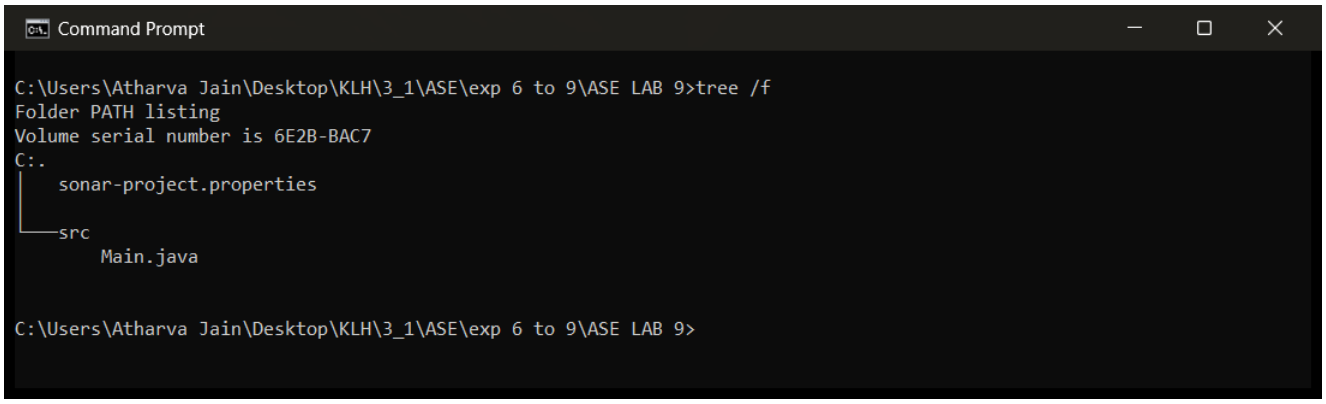
Step 4: Create the Java Project Folder

Create/open the Lab 9 folder:

```
C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9\ASE LAB 9
```

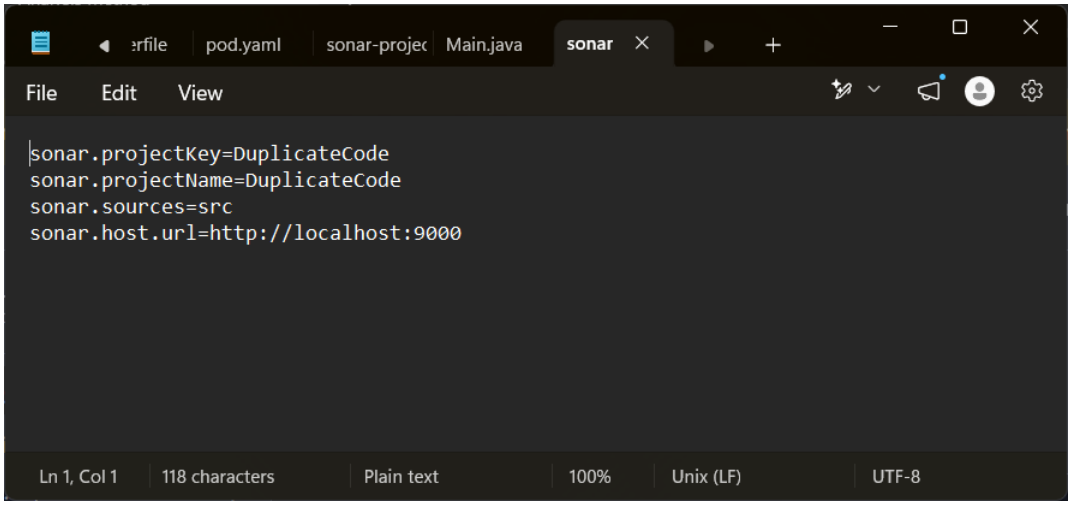
The folder structure should be:

```
ASE LAB 9
│
├── sonar-project.properties
│
└── src
    └── Main.java
```



Create/open sonar-project.properties and add:

```
sonar.projectKey=DuplicateCode
sonar.projectName=DuplicateCode
sonar.sources=src
sonar.host.url=http://localhost:9000
```



Open:

```
C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9\ASE LAB 9\src\Main.java
```

Use the following code. The same large block is intentionally repeated in two methods so SonarQube can detect duplicate code:

```
public class Main {
```

```
    public void calculateSalary() {

        int basic = 30000;
        int hra = 5000;
        int da = 3000;
        int ta = 2000;
        int medical = 1500;

        int total = basic + hra + da + ta + medical;

        System.out.println("Basic: " + basic);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("TA: " + ta);
        System.out.println("Medical: " + medical);
        System.out.println("Total: " + total);
        System.out.println("Salary calculated");
        System.out.println("Employee eligible");
        System.out.println("Salary processed");
    }

    public void calculateBonus() {

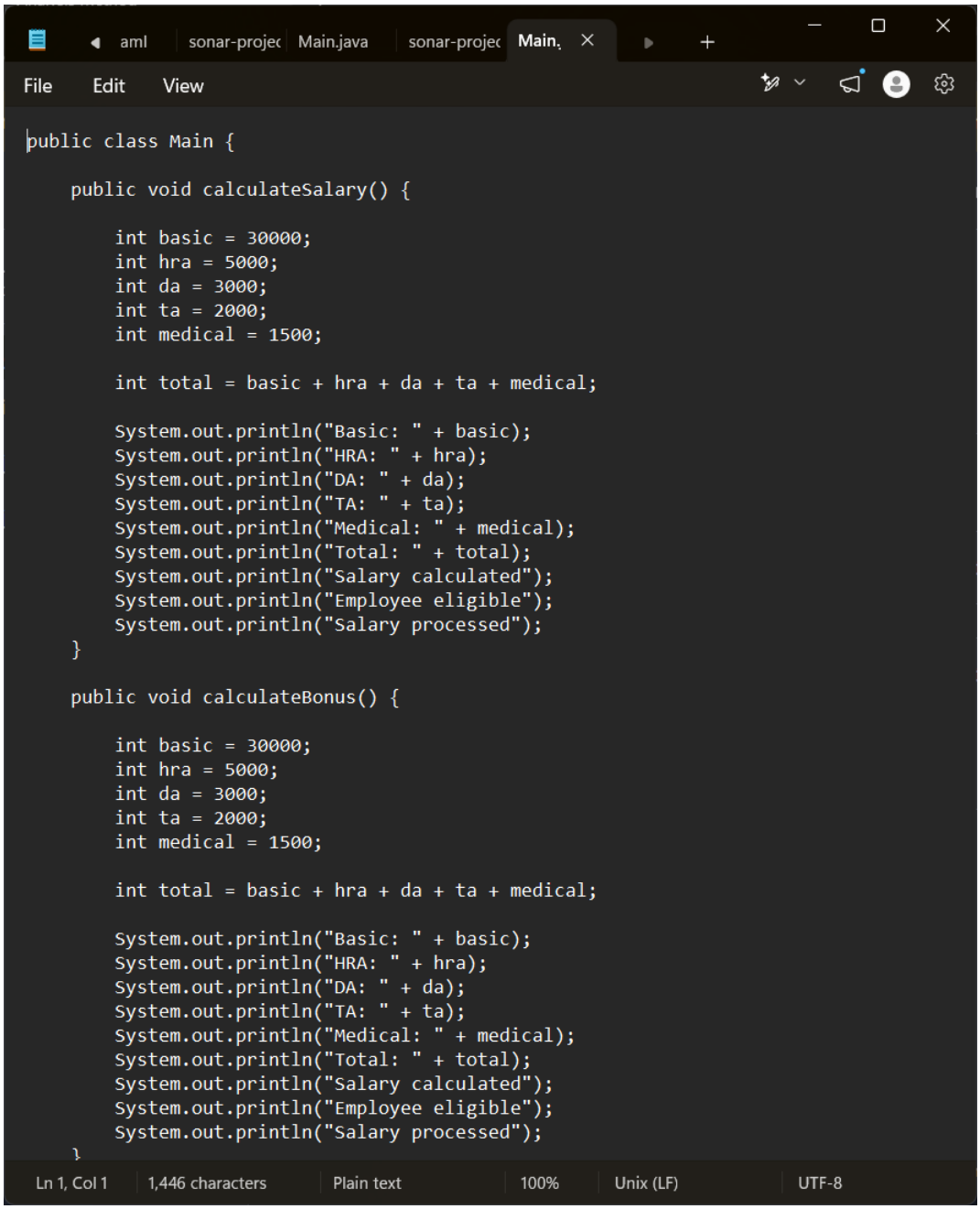
        int basic = 30000;
        int hra = 5000;
        int da = 3000;
        int ta = 2000;
        int medical = 1500;

        int total = basic + hra + da + ta + medical;

        System.out.println("Basic: " + basic);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("TA: " + ta);
        System.out.println("Medical: " + medical);
        System.out.println("Total: " + total);
        System.out.println("Salary calculated");
        System.out.println("Employee eligible");
        System.out.println("Salary processed");
    }

    public static void main(String[] args) {
        Main obj = new Main();

        obj.calculateSalary();
        obj.calculateBonus();
    }
}
```



The code in calculateSalary() and calculateBonus() contains the same repeated block. This is the duplicate code that SonarQube will analyze.

Step 5: Install / Check SonarScanner

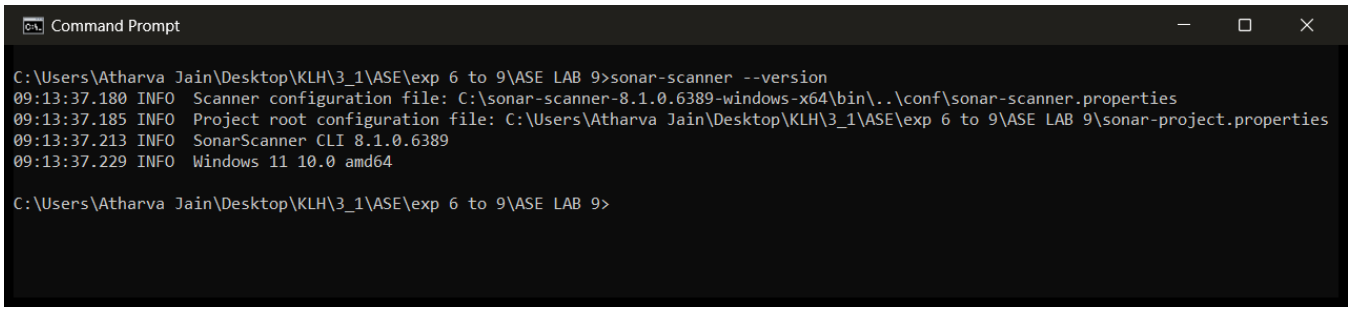
SonarScanner is used to send the Java project to SonarQube for analysis.

Check whether SonarScanner is installed:

```
sonar-scanner --version
```

You should see the SonarScanner version, for example:

```
SonarScanner CLI 8.1.0.6389
```



Step 6: Generate SonarQube Authentication Token

Go to SonarQube and login.

Then:

1. Open your profile from the top-right corner.

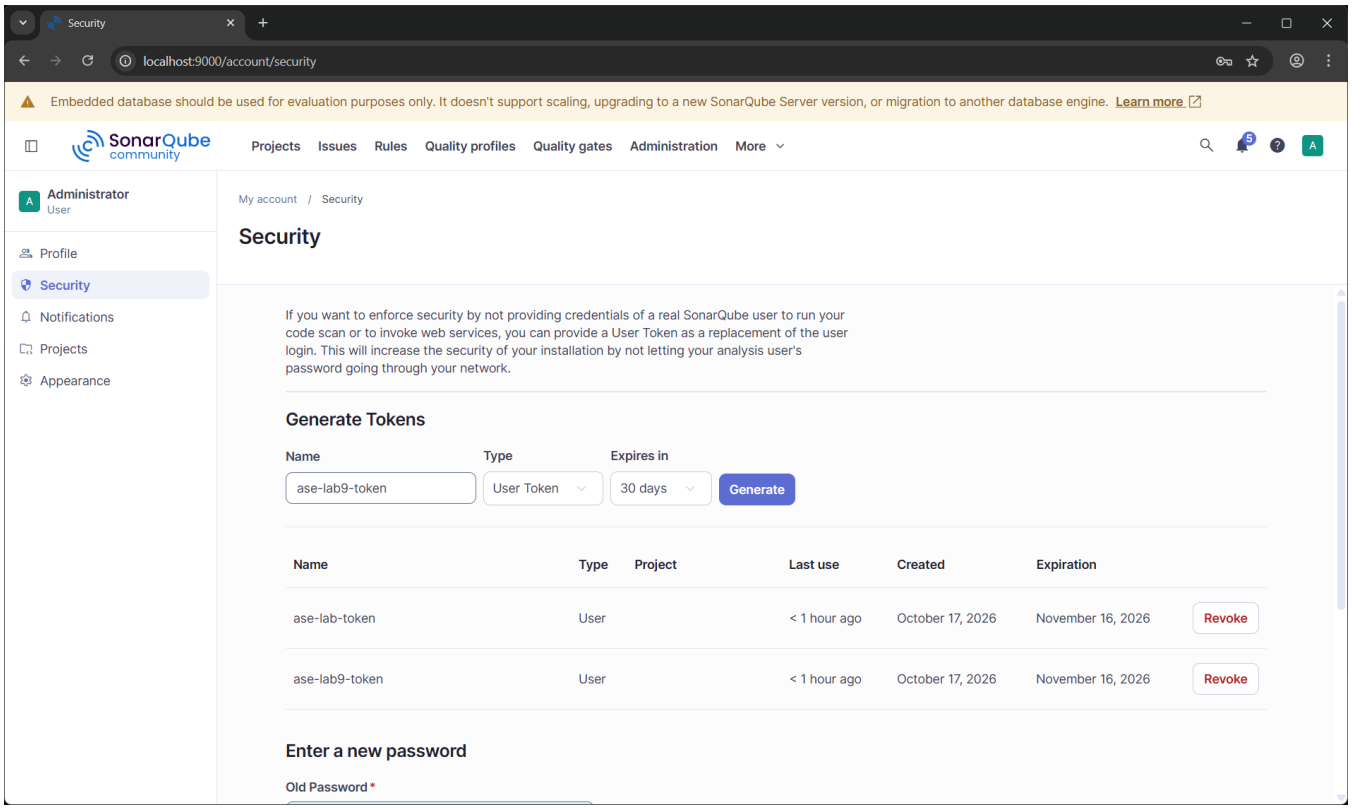
2. Open Security.

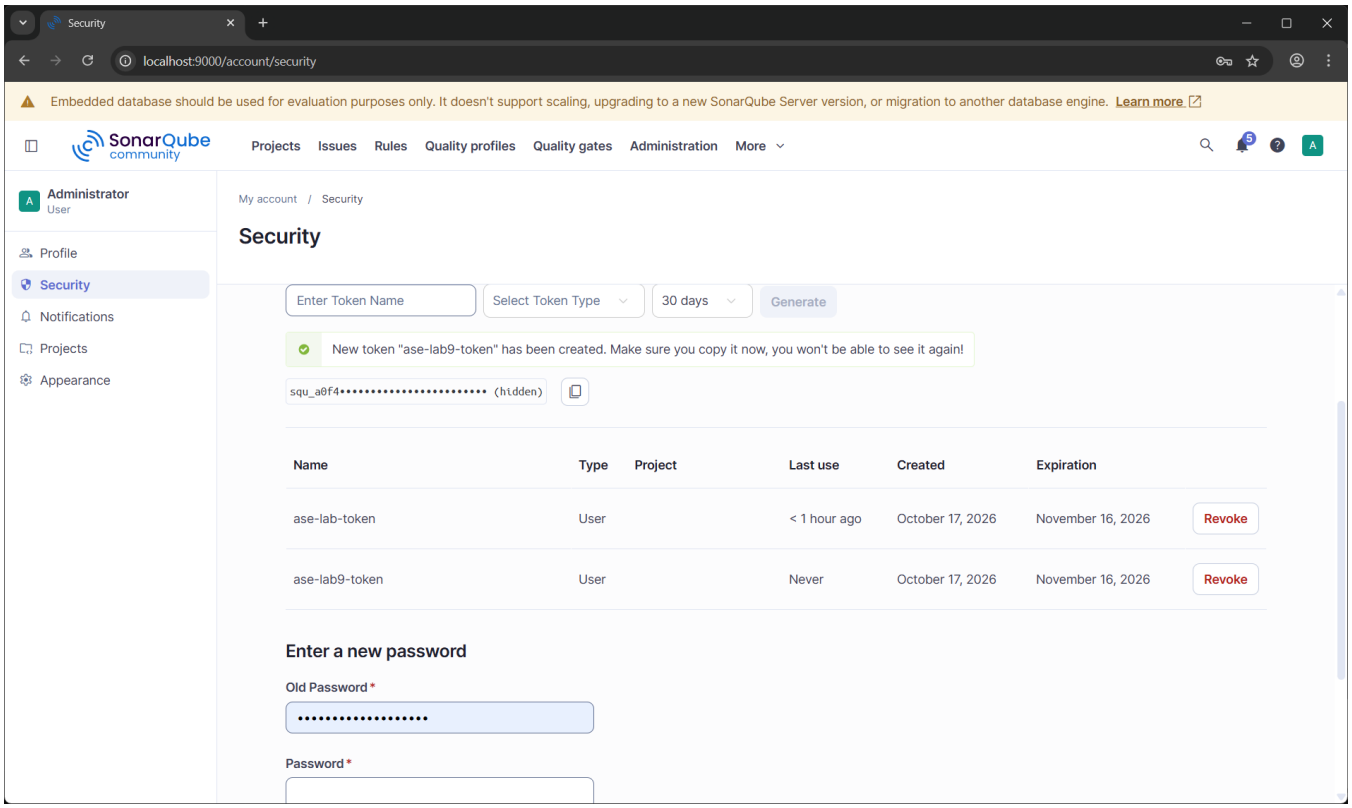
3. Enter a name for the token.

4. Select User Token if that option is shown.

5. Generate the token.

6. Copy the token and keep it safe. Do not share it.





Open Command Prompt and set the token for the current CMD window:

```
set SONAR_TOKEN=YOUR_TOKEN
```

Replace YOUR_TOKEN with the token you generated.

Step 7: Go to the Project Directory

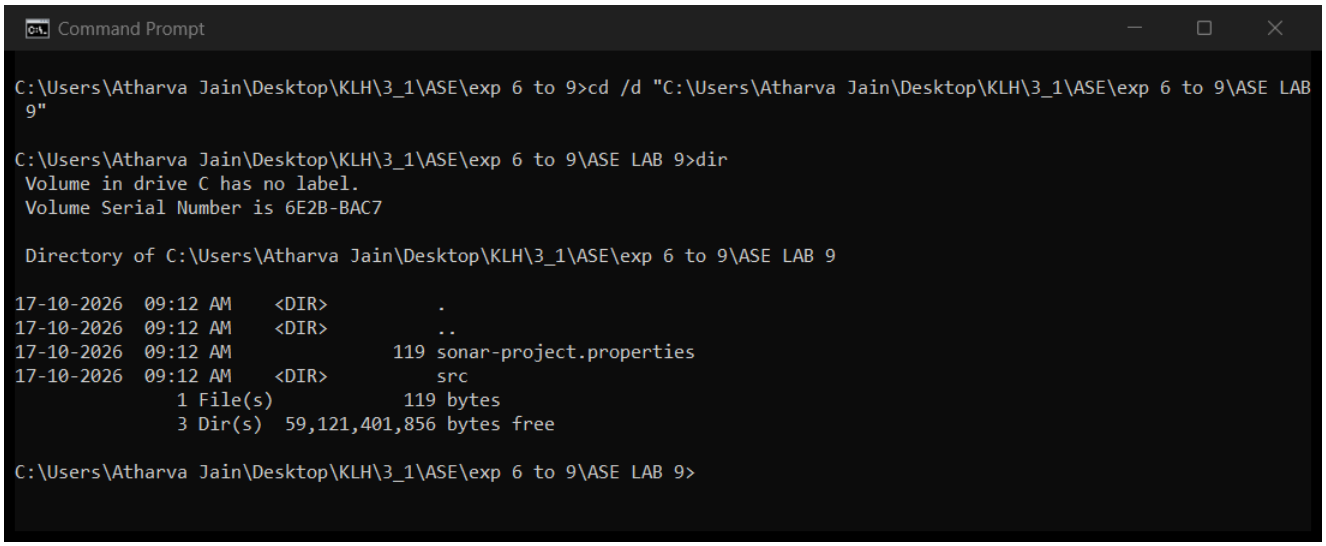
Open Command Prompt and move to the Lab 9 project folder:

```
cd /d "C:\Users\Atharva Jain\Desktop\KLH\3_1\ASE\exp 6 to 9\ASE LAB 9"
```

Check the files:

```
dir
```

You should see sonar-project.properties and the src folder.



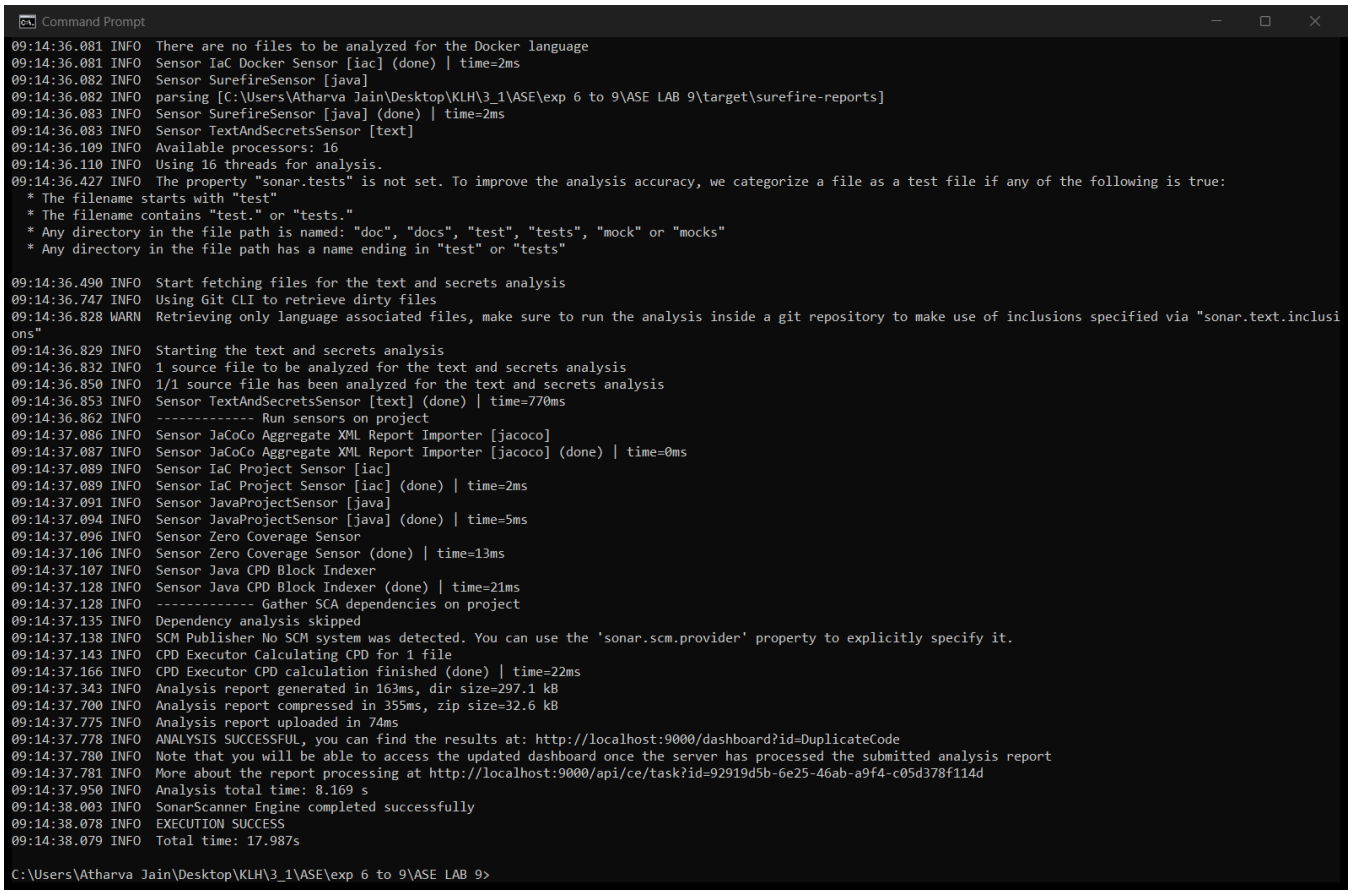
Step 8: Run SonarScanner

Run:

```
sonar-scanner
```

Wait until the analysis is completed. The scanner should finish with:

```
EXECUTION SUCCESS
```



If you get an old .scannerwork permission error, run:

```
rmdir /s /q ".scannerwork"
```

Then run again:

```
sonar-scanner
```

If you get HTTP 401 Unauthorized, set SONAR_TOKEN again in the same CMD window and run sonar-scanner again.

Step 9: Open the SonarQube Project Dashboard

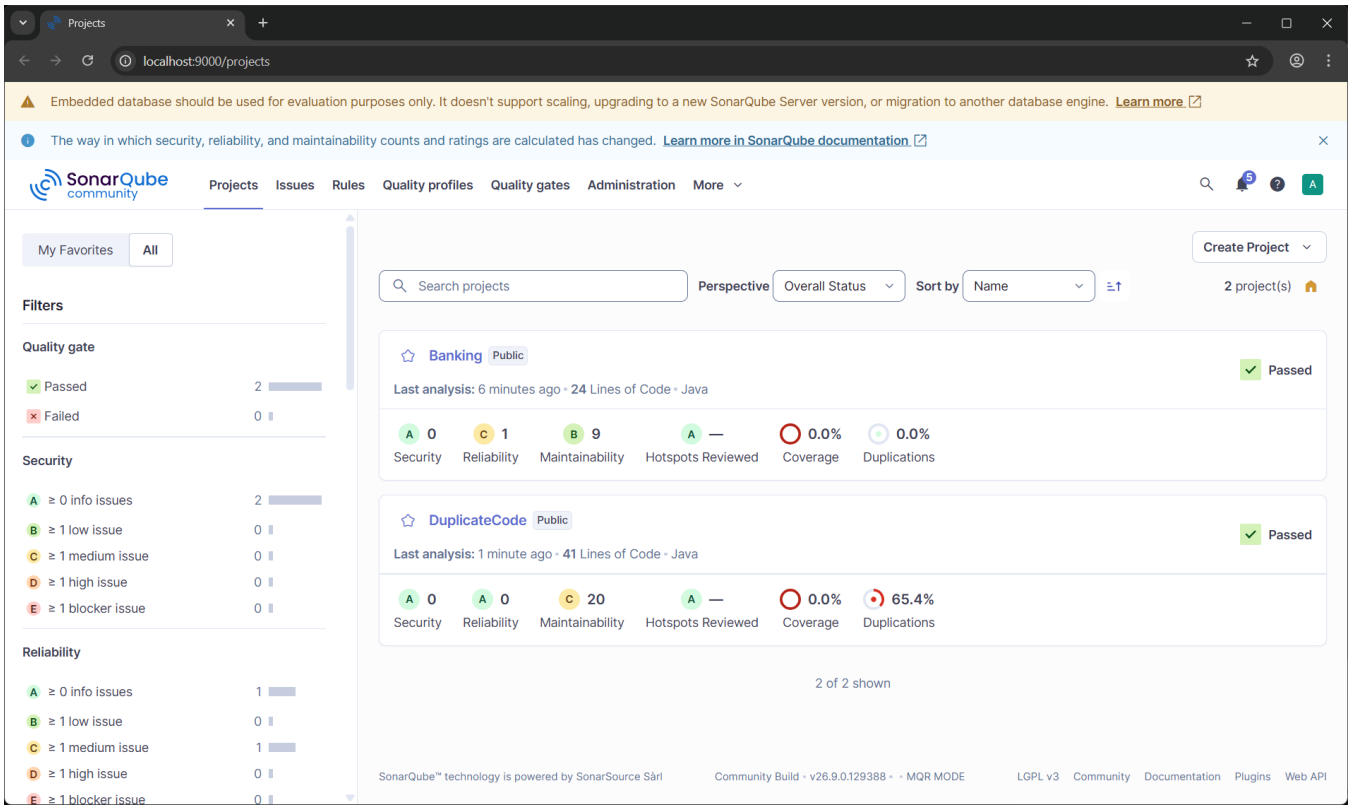
Open:

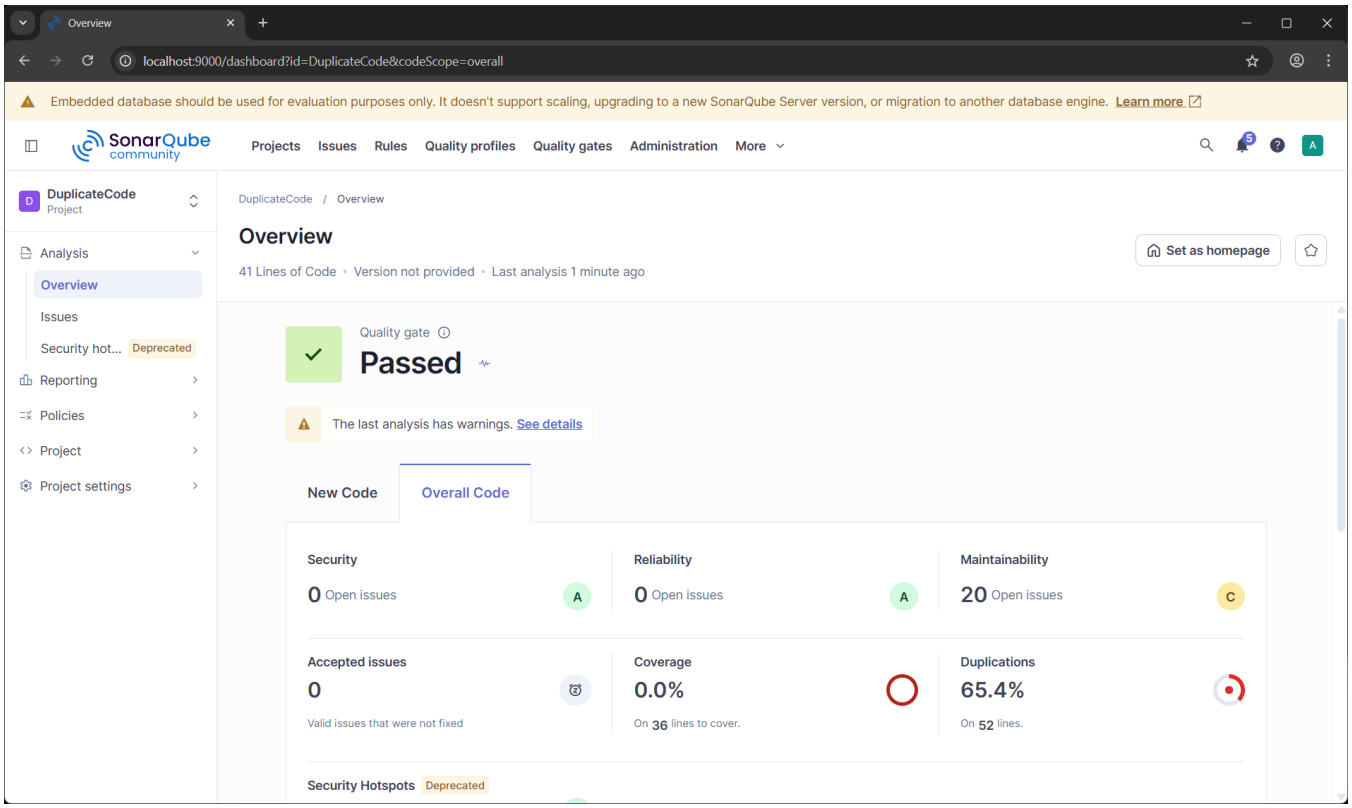
```
http://localhost:9000
```

Then go to:

Projects → DuplicateCode

Open the project Overview.





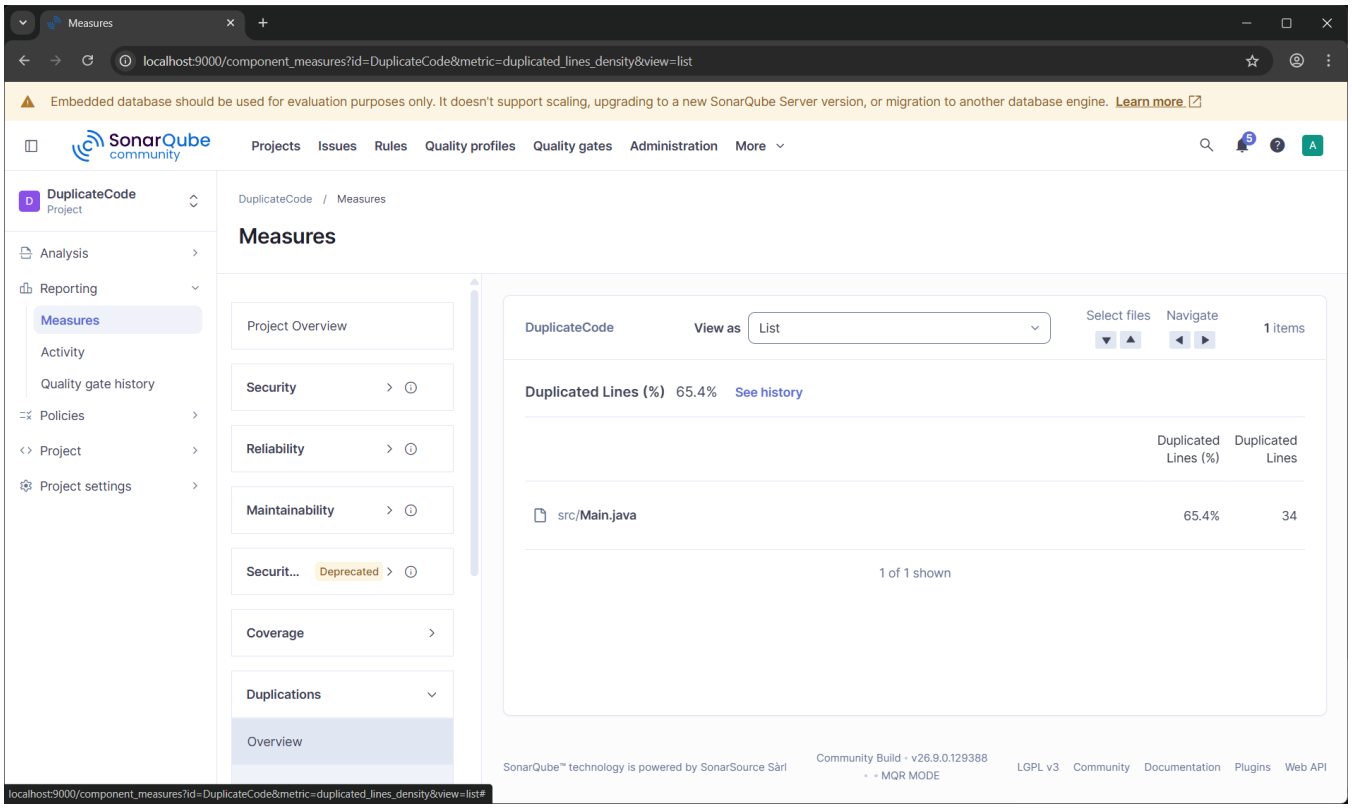
Step 10: Check Duplicate Code

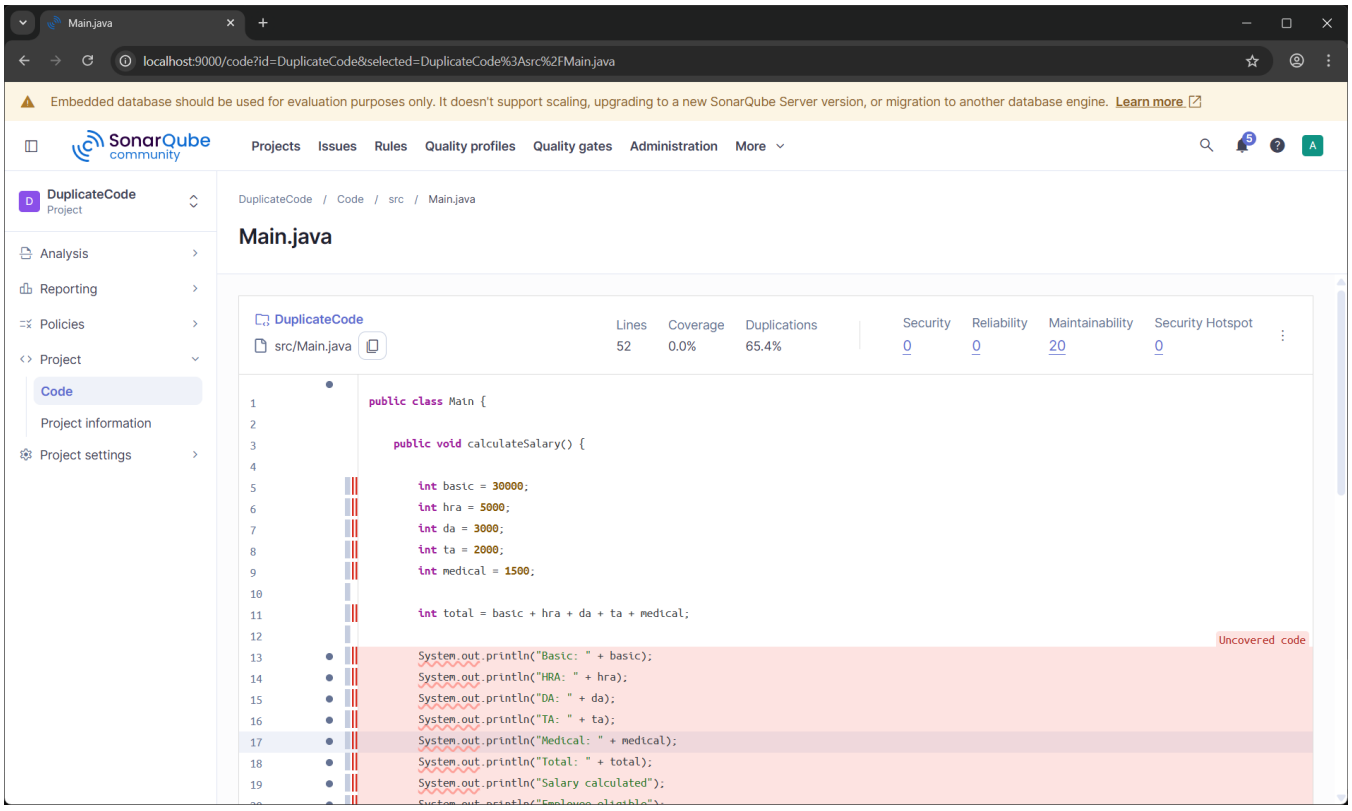
In the dashboard, check the Duplications metric.

For this experiment, SonarQube should identify the repeated code and show a non-zero Duplications percentage.

The exact percentage depends on the code and SonarQube's CPD analysis. Record the percentage shown on your own dashboard.

Recorded on this system: Duplications = 65.4% (on 52 lines) for src/Main.java. The repeated block in calculateSalary() and calculateBonus() is highlighted as duplicated code.





What is CPD?

CPD means Copy-Paste Detection. It is the SonarQube analysis used to find sufficiently large repeated code blocks. The Duplications metric shows the amount of duplicated code detected.

Observation

The Java program containing repeated code was analyzed using SonarQube. SonarQube's Copy-Paste Detection identified the repeated code, and the Duplications metric was displayed in the project dashboard.

Result

Thus, duplicate code was successfully analyzed and detected using SonarQube.

Screenshots to Take

- Docker Desktop / Command Prompt showing the SonarQube container running.
- SonarQube DuplicateCode project page.
- sonar-project.properties file.
- Main.java showing the repeated code.
- Command Prompt showing sonar-scanner and EXECUTION SUCCESS.
- Final DuplicateCode dashboard showing the Duplications percentage.

Important Notes

- Do not share your SonarQube authentication token.
- The exact Duplications percentage can be different for different code.
- A Quality Gate can be Failed because of other conditions such as coverage or issues; this is separate from whether duplicate code was detected.
- An SCM provider warning does not by itself mean duplicate-code analysis failed.